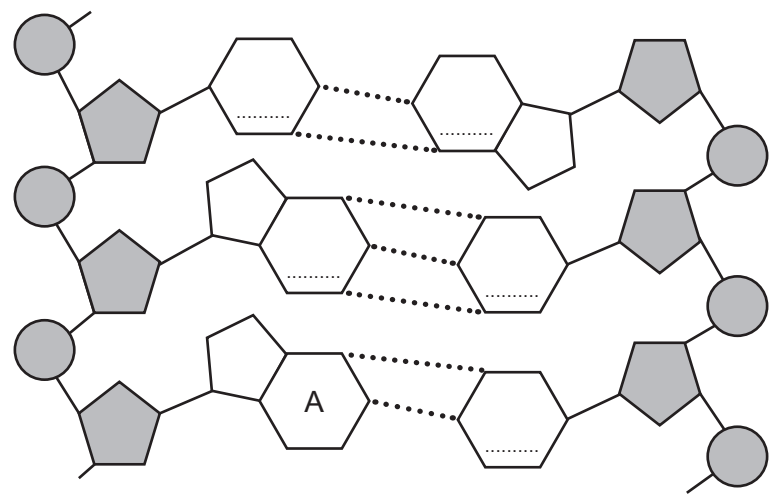


5. (a) **Image 5.1** represents part of a DNA molecule.

Image 5.1



- (i) Circle **one** nucleotide in **Image 5.1**. [1]
- (ii) Using the letters **A, C, T, G**, label **all** the organic bases in **Image 5.1**. One adenine is labelled already. [2]
- (iii) In the late 1940s, Erwin Chargaff published his research into the base composition of the DNA in different species. Some of his findings are shown in **Table 5.2**.

Table 5.2

Organism	Percentage of base			
	Adenine	Guanine	Cytosine	Thymine
maize	26.8	22.8	23.2	27.2
chicken	28.0	22.0	21.6	28.4
yeast	31.3	18.7	17.1	32.9
sea urchin	32.8			

I. Complete **Table 5.2** to suggest values for the sea urchin. [1]



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II. Based on the results in **Table 5.2**, state the conclusion that can be made regarding the base composition of DNA. Explain your answer. [3]

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(b) Meselson and Stahl also carried out investigations on DNA in the 1950s. They proposed the theory of semi-conservative replication. State what is meant by semi-conservative replication. [2]

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(c) DNA contains only four different bases, but proteins can be made of up to 20 different amino acids. Explain how these four bases allow for the production of so many different proteins. [4]

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